

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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1. Facility Name

Calvert Cliffs Nuclear Power Plant, Unit 1

☒ **050**
☐ **052****2. Docket Number**

05000317

3. Page

1 OF 4

4. Title

Control Room Loss of Ventilation due to Smoke Detector Failure Causing Common Supply Damper to Shut

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☒ 050	Docket Number
07	31	2025	2025	- 001 -	01	11	19	2025	Calvert Cliffs Nuclear Power Plant, Unit 2	☐ 052	05000318

9. Operating Mode

1 – Power Operation

10. Power Level

100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☒ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).**12. Licensee Contact for this LER****Licensee Contact**

Travis E. Lefton, Principal Regulatory Engineer

Phone Number (Include area code)

667-313-6402

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	VI	DET	P435	Y	N/A				

14. Supplemental Report Expected☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)**15. Expected Submission Date**

Month Day Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On July 31, 2025, at 1744, the Main Control Room (MCR), shared by Units 1 and 2, experienced a loss of ventilation due to the Control Room supply damper fully closing. Closure of the damper isolated the ventilation flow to the Control Room which resulted in both trains of the Control Room Emergency Ventilation System and Control Room Emergency Temperature System being inoperable, resulting in Technical Specification Limiting Conditions for Operation (LCOs) 3.7.8 and 3.7.9 not being met, respectively, for Units 1 and 2. At time 1745, the Units exited the associated LCOs after the damper returned to its normal, open position. Field troubleshooting isolated the cause of the intermittent closure of the damper to a MCR ventilation smoke detector. The smoke detector functions to shut the MCR supply damper by energizing the associated solenoid valve. This event is reportable in accordance with 10 CFR 50.73(a)(2)(v)(D) and 10 CFR 50.73(a)(2)(vii). The direct cause of the smoke detector failure was an electrical short, due to condensation, on the control board of the Control Room Air Conditioning (A/C) return duct smoke detector (0AE5347) that sends an automatic closure signal to the Control Room A/C dampers.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME Calvert Cliffs Nuclear Power Plant, Unit 1	☒ 050	2. DOCKET NUMBER 05000317	3. LER NUMBER		
	☐ 052		YEAR 2025	SEQUENTIAL NUMBER - 001	REV NO. - 01

NARRATIVE**PLANT AND SYSTEM IDENTIFICATION**

Calvert Cliffs Nuclear Power Plant, Units 1 and 2, are Combustion Engineering Pressurized Water Reactors with a licensed maximum power level of 2737 megawatts thermal. The Energy Industry Identification System code used in the text is identified as [VI].

A. CONDITION PRIOR TO EVENT

Units: 1/2
Date: July 31, 2025
Power Level: 100/100 percent
Mode: MODE 1/1

B. DESCRIPTION OF EVENT

On July 31, 2025, at 1744, the Main Control Room (MCR), shared by Units 1 and 2, experienced a loss of ventilation due to the MCR supply damper fully closing. Closure of the damper isolated the ventilation flow to the MCR which resulted in both trains of the Control Room Emergency Ventilation System (CREVS) and Control Room Emergency Temperature System (CRETS) being inoperable, resulting in Technical Specification (TS) Limiting Conditions for Operation (LCO) 3.7.8 and 3.7.9 not being met, respectively, for Units 1 and 2. At time 1745 the Units exited the associated LCOs after the damper returned to its normal, open position.

C. DATES AND APPROXIMATE TIMES OF MAJOR OCCURRENCES

July 31, 2025

- 17:44:00 Unit 1 entered LCOs 3.0.3, 3.7.8, and 3.7.9 due to ventilation damper 0-CV-5342 closing rendering two CREVS trains inoperable in MODE 1.
- 17:44:00 Unit 2 entered LCO 3.0.3, 3.7.8, and 3.7.9 due to ventilation damper 0-CV-5342 closing rendering two CRETS trains inoperable in MODE 1.
- 17:44:30 Ventilation damper 0-CV-5342 re-opened. (time is approximate)
- 17:45:00 Unit 1 exited 3.0.3, 3.7.8, and 3.7.9 due to damper 0-CV-5342 returned to its normal open position. Both Trains of CREVS and CRETS were restored to service.
- 17:45:00 Unit 2 exited 3.0.3, 3.7.8, and 3.7.9 due to damper 0-CV-5342 returned to its normal open position. Both Trains of CREVS and CRETS were restored to service.

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NARRATIVE**D. CAUSE OF EVENT**

The direct cause of the smoke detector failure was an electrical short, due to condensation, on the control board of the Control Room Air Conditioning (A/C) return duct smoke detector (0AE5347) that sends an automatic closure signal to the Control Room A/C dampers.

A cause that led to the event was procedure use and adherence by engineering personnel. The previous July 2018 occurrence resulted in actions to address the automatic function of the damper. Site procedures were not adhered to which resulted in untimely component Life Cycle Management process completion and inadequate closure of risk elimination actions. Key information was not entered into approved Risk Management and System Health management software. The software tool is periodically reviewed by senior leadership. The review would have provided the opportunity to increase priority of open actions.

E. SAFETY ANALYSISSystem Design

The Control Room Envelope air conditioning system is divided into three supply and return duct systems: one for each Cable Spreading Room and one for the Main Control Room. A portion of supply and return air is also routed to the Control Room heating, ventilation, and air conditioning (HVAC) equipment room. Each supply and return branch contains an isolation damper. Smoke detectors are located in the return duct from each zone. In the event of a fire in one zone, the smoke detector automatically closes the corresponding isolation dampers. The air conditioning system will then continue to serve the other two zones without interruption. The Control Room HVAC Supply Damper (0-CV-5342) shuts on actuation of a smoke detector (0AE5347) or the operation of the associated hand-switch to CLOSE. There was no fire alarm at the time the damper was observed in the closed position, and the Control Room HVAC Supply Damper Hand Switch (0-HS-5342) was in the OPEN position.

Both trains of CREVS and CRETS being inoperable satisfies the criteria in NUREG-1022, Revision 3, for any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident and a common cause of inoperability. Therefore, this event is being reported pursuant to 10 CFR 50.73(a)(2)(v)(D) and 50.73(a)(2)(vii).

No structures, systems, or components that were inoperable at the start of the event contributed to the event. The safety significance of the event was minimal due to the short duration of the event. MCR temperature and habitability were maintained during the event.

F. CORRECTIVE ACTIONS**Action(s) Completed:**

- The site has completed a permanent Engineering Change and associated 50.59 evaluation to permanently remove the automatic function of the smoke detector to close the MCR HVAC damper and credit a manual operator action to meet the credited fire safety function.

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	☐ 052		<table border="1"><thead><tr><th>YEAR</th><th>SEQUENTIAL NUMBER</th><th>REV NO.</th></tr></thead><tbody><tr><td>2025</td><td>- 001</td><td>- 01</td></tr></tbody></table>	YEAR	SEQUENTIAL NUMBER	REV NO.	2025
YEAR	SEQUENTIAL NUMBER	REV NO.					
2025	- 001	- 01					

NARRATIVE**Actions in progress:**

- Complete configuration change to permanently remove the automatic function of the smoke detector to close the MCR HVAC damper.
- Update procedures and alarm manual response to support manual operation of the MCR HVAC damper

G. PREVIOUS OCCURRENCES

A similar issue occurred in July 2018 as documented in LER 317-2018-003-01.

H. COMPONENT FAILURE DATA

Component IEEE 803	IEEE805
FUNCTION ID	SYSTEM ID
Detector DET	VI
Manufacturer Pyrotechnics	